

Protein Structural Comparison and Classification

using a Linear Algebra approach

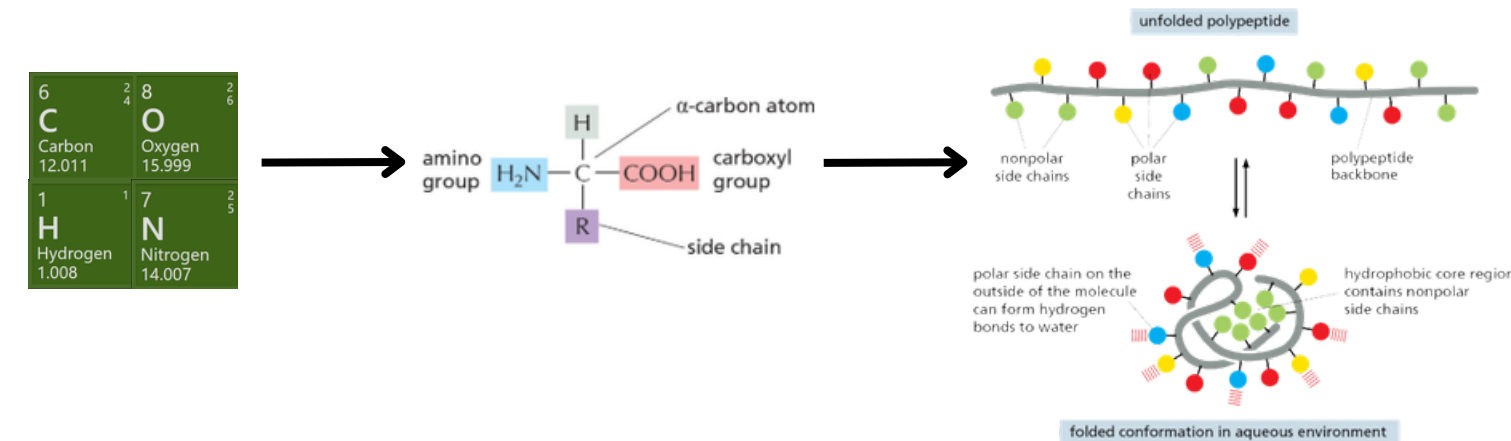
1. Question of the day

Biological life largely depends on protein interactions, which are governed by protein structure. Classifying proteins by structure helps predict their function—key to crops, chemical synthesis, biofuels and more importantly, drug development.

How can we compare and classify new proteins using those with known structures?

3. Protein and Hydrophobic Interactions

Proteins are complex substances present in all living organisms, composed of amino acids.



The function of a protein depends on its structure and folding patterns. One of the main forces helping it folds is called the hydrophobic interaction. This happens because some parts of a protein don't mix well with water (just like oil doesn't mix with water). These "water-fearing" parts (called hydrophobic) tend to cluster together on the inside of the protein, away from water, while the "water-loving" parts stay on the outside. This helps the protein stay stable and do its job.

4. Mathematical Approach

We first build contact maps for each protein based on residue proximity, flatten them into vectors, and assemble these into a contact matrix (details in section 5). Singular Value Decomposition (SVD) is then used to reduce dimensionality and extract structural patterns for efficient protein comparison. This section presents concepts of SVD.

$$A = D S T^T$$

Given an $m \times n$ matrix of documents and terms, A , and rank $(A) = r \leq \min(m, n)$, the singular value decomposition of A is defined by the formula: $A = D S T^T$. Where $D^T D = T^T T = I_n$ and $S = \text{diag}(\sigma_1, \dots, \sigma_r)$, $\sigma_i > 0$ for $1 \leq i \leq r$, $\sigma_j = 0$ for $j \geq r + 1$. The r columns of the orthogonal matrices D and T define the orthonormal eigenvectors associated with the r nonzero eigenvalues of respectively $A A^T$ and $A^T A$. The matrix A can be approximated in A^k , by modifying the three factored matrices. From S matrix, the k largest singular values may be kept, and the remaining smaller ones set to zero. To obtain the new matrices D^k and T^k keep the first k columns of the corresponding matrices D and V .

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2. Problem Statement

This project aims to compare and classify protein structures into fold families based on pattern of hydrophobic intrachain interactions using Singular Value Decomposition (SVD).

5. Implementation

5.1.Database selection

All proteins used in this project are retrieved from the RCSB Protein Data Bank. Our sample dataset comprises 5 different myoglobin proteins that serve as the benchmark. 2 non-myoglobin proteins are randomly recruited for comparison and classification.

5.2.Contact map

A binary matrix is created based on the proximity of 2 residues. If their distance is $< 7\text{\AA}$, they are interacting and are given 1, otherwise 0.

5.3.Contact vector

The contact map is then linearized to construct the contact vector. The i th entry of this vector is calculated by $\mathbf{i} = (\mathbf{x} - \mathbf{1})\mathbf{b} + \mathbf{y}$, where $\mathbf{b}$ is the number of residues in this protein.

$$\begin{bmatrix} 0 & 1 & 0 & \dots & 0 & 0 & 0 \\ 1 & 0 & 1 & \dots & 0 & 0 & 0 \\ 0 & 1 & 0 & \dots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & 1 & 0 \\ 0 & 0 & 0 & \dots & 1 & 0 & 1 \\ 0 & 0 & 0 & \dots & 0 & 1 & 0 \end{bmatrix}$$

$$[0 \ 1 \ 0 \ \dots \ 0 \ 1 \ 0]$$

5.4.Contact matrix

As different proteins have different amounts of residues, the length of their contact vectors is also different. To make all the contact vectors the same size, we take the size of the largest contact vector $\rightarrow$ normalize all the other contact vectors to this size (padding '0') $\rightarrow$ group them all together and obtain an $n \times m$ matrix, where n is the number of proteins of the database and m is the size of the largest contact vector

5.5.Contact matrix SVD

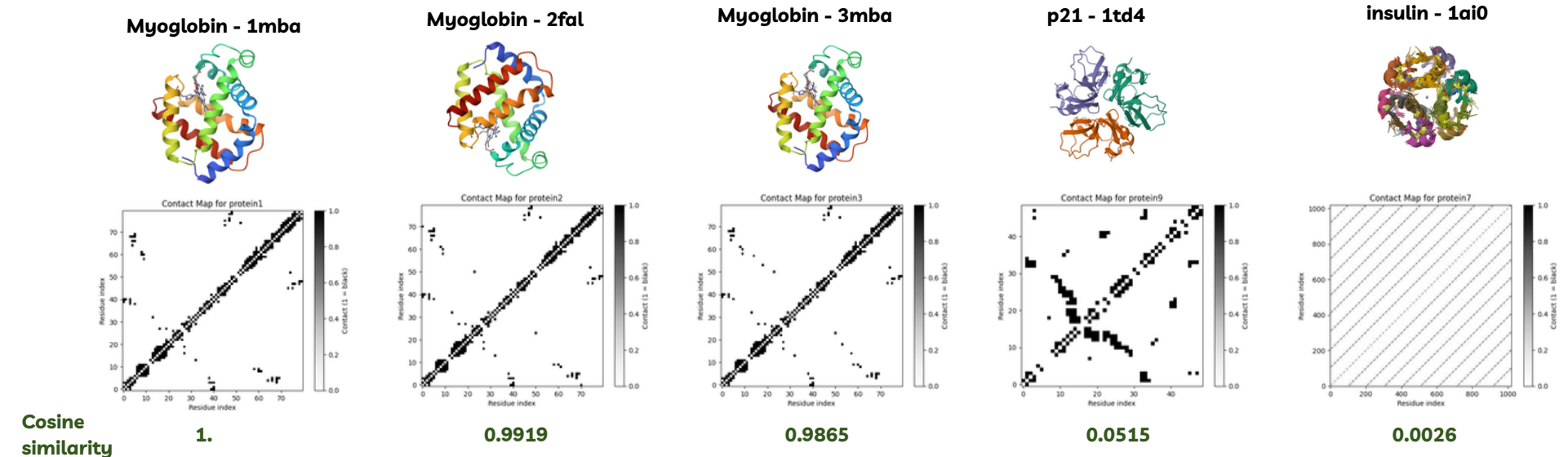
Singular Value Decomposition (SVD) is applied to the contact matrix to extract major structural patterns and reduce noise. This helps map proteins into a lower-dimensional space for efficient comparison.

5.6.Cosine similarity

Proteins are compared using cosine similarity between their reduced-dimensional vectors, allowing classification into fold families or detection of structural similarity. Q is the protein query, D is the document vector.

$$\cos \theta = \frac{Q \cdot D}{\|Q\| \|D\|}$$

6. Results



7. Limitation

- The choice of the value of k is challenging because it is directly related to dimensionality reduction. It must be a value that captures all the true structures in the data while remaining small enough to exclude noise and redundant information from the new representation.
- Due to computational limits, we only tested our model with proteins that have small residues. Cosine similarity was calculated using the first indexed protein of the known family. Further analysis across the full dataset is needed to validate this approach.